

Chemical Engineering Thermodynamics

Assignment 1

Katerina Bompai, Bushra Alkarmi

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Abstract

This report addresses two problems. In problem A, we model an ideal Brayton (gas-turbine) cycle with isentropic compression/expansion and isobaric heat addition, deriving expressions for compressor work, turbine work, net work, heat addition, and thermal efficiency from ideal-gas relations and steady-flow energy balances. The report then identifies the pressure ratio P_2/P_1 that maximizes net power; computes the minimum air flow (MMSCFD) required to produce 100 MW; reports the cycle efficiency at maximum-power conditions; determines the P_2/P_1 that maximizes efficiency and the corresponding maximum efficiency; evaluates the net power at maximum-efficiency conditions; compares performance to the Carnot efficiency; and estimates the kerosene requirement (lb/h) at maximum power using its lower heating value. Assumptions and conversions are summarized in the text; numerical results appear in the Results section.

In Problem B, we compute the vapor fraction α of a two-phase, multi-component hydrocarbon mixture. We derive the analytical derivative and solve for $\alpha \in [0, 1]$ with both Newton's method and the Secant method, provide iteration tables, and confirm the root using Excel Goal Seek. The function `SUMPRODUCT` is used for compact evaluations of the objective and its derivative. Numerical values and composition data are reported in the Results section.¹

¹LaTeX coding/formatting was assisted by ChatGPT (OpenAI); Results were produced and verified by the authors.

1 Problem A: Ideal Brayton Cycle Analysis

1.1 Known Data and Assumptions

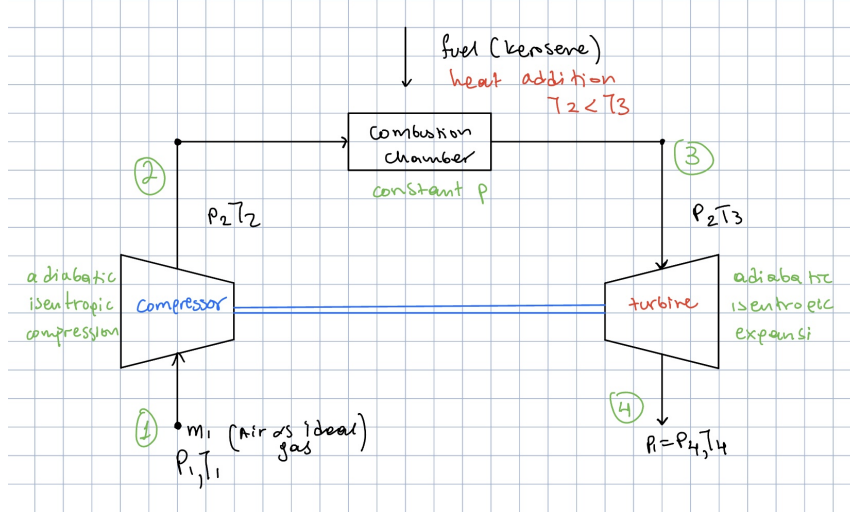


Figure 1: Brayton cycle schematic.

State labeling

(1) compressor inlet, (2) compressor exit, (3) turbine inlet, (4) turbine exit.

Given data & constants

Table 1: Problem A inputs and constants.

Quantity	Value
Inlet temperature, T_1	303.15 K
Maximum cycle temperature, T_3	1523.15 K
Heat capacity at constant pressure, c_p (air)	$1.004 \text{ kJ kg}^{-1} \text{ K}^{-1}$
Heat capacity ratio, γ (air)	1.4
Specific gas constant, R (air)	$0.287 \text{ kJ kg}^{-1} \text{ K}^{-1}$
Target net power for sizing, $\dot{W}_{\text{net}}$	100 MW
Kerosene lower heating value (LHV)	43 MJ kg^{-1}
Standard conditions for SCF [†]	$P_{\text{std}} = 1 \text{ atm}, T_{\text{std}} = 288.706 \text{ K} (60^\circ \text{F})$

Modeling assumptions

- Steady state, steady flow; kinetic/potential energy changes are negligible.

- Working fluid: air as an ideal gas with constant properties c_p , γ .
- Isentropic stages: $s_2 = s_1$ across the compressor and $s_4 = s_3$ across the turbine (adiabatic, reversible).
- Heat addition (2→3) is **isobaric**; component pressure losses neglected.
- Same pressure ratio across compressor and turbine: $P_2/P_1 = P_3/P_4 = r_p$.

1.2 State Relations (Isentropic)

For an ideal gas with constant specific heats,

$$\gamma \equiv \frac{c_p}{c_v},$$

and an isentropic (adiabatic, reversible) stage satisfies $Pv^\gamma = \text{const.}$ Using the ideal-gas law $Pv = RT$.

Compressor (1 → 2). (1) Isentropic relation

$$\boxed{P_1 v_1^\gamma = P_2 v_2^\gamma}. \quad (1)$$

(2) Ideal-gas law at each state

$$P_1 v_1 = RT_1, \quad P_2 v_2 = RT_2 \implies \boxed{\frac{v_2}{v_1} = \frac{T_2}{T_1} \frac{P_1}{P_2}}. \quad (2)$$

(3) Combine (1) and (2) Divide (1) by $P_1 v_1^\gamma$ and substitute (2):

$$\begin{aligned} 1 &= \frac{P_2}{P_1} \left(\frac{v_2}{v_1} \right)^\gamma = \frac{P_2}{P_1} \left[\frac{T_2}{T_1} \frac{P_1}{P_2} \right]^\gamma \\ &= \frac{P_2}{P_1} \left(\frac{T_2}{T_1} \right)^\gamma \left(\frac{P_1}{P_2} \right)^\gamma \quad (\text{distribute the power over the product}) \\ &= \left(\frac{T_2}{T_1} \right)^\gamma \underbrace{\left[\frac{P_2}{P_1} \left(\frac{P_1}{P_2} \right)^\gamma \right]}_{= \left(\frac{P_1}{P_2} \right)^{\gamma-1}}. \end{aligned}$$

Hence

$$\left(\frac{T_2}{T_1} \right)^\gamma = \left(\frac{P_2}{P_1} \right)^{\gamma-1}.$$

Let $r \equiv P_2/P_1$ and $a \equiv \frac{\gamma-1}{\gamma}$. Taking the $1/\gamma$ power:

$$\boxed{\frac{T_2}{T_1} = r^{\frac{\gamma-1}{\gamma}}}$$

Turbine (3 → 4): same steps. Isentropic: $P_3 v_3^\gamma = P_4 v_4^\gamma$. From the ideal-gas law,

$$\frac{v_4}{v_3} = \frac{T_4}{T_3} \frac{P_3}{P_4}.$$

Repeat the algebra:

$$1 = \frac{P_4}{P_3} \left(\frac{v_4}{v_3} \right)^\gamma = \left(\frac{T_4}{T_3} \right)^\gamma \left(\frac{P_3}{P_4} \right)^{\gamma-1} \Rightarrow \left(\frac{T_3}{T_4} \right)^\gamma = \left(\frac{P_3}{P_4} \right)^{\gamma-1}.$$

With $P_3/P_4 = r$,

$$\boxed{\frac{T_3}{T_4} = r^{\frac{\gamma-1}{\gamma}} \Rightarrow T_4 = T_3 r^{-\frac{\gamma-1}{\gamma}}}.$$

1.3 Power for Compressor and Turbine

Specific work (per unit mass of air).

$$w_c = c_p (T_2 - T_1), \quad (1)$$

$$w_t = c_p (T_4 - T_3), \quad (2)$$

$$w_{\text{net}} = w_c + w_t = c_p [(T_3 - T_4) - (T_2 - T_1)]. \quad (3)$$

Power rates (with mass flow $\dot{m}$).

$$\dot{W}_c = \dot{m} w_c, \quad \dot{W}_t = \dot{m} w_t, \quad \dot{W}_{\text{net}} = \dot{m} w_{\text{net}}. \quad (4)$$

In terms of pressure ratio $r \equiv P_2/P_1$. Let $a \equiv \frac{\gamma-1}{\gamma}$, so $T_2 = T_1 r^a$ and $T_4 = T_3 r^{-a}$. Then

$$w_c(r) = c_p (T_1 r^a - T_1), \quad (5)$$

$$w_t(r) = c_p (T_3 - T_3 r^{-a}), \quad (6)$$

$$w_{\text{net}}(r) = c_p [(T_3 - T_1) - T_3 r^{-a} - T_1 r^a]. \quad (7)$$

1.4 Optimize Net Work: Solve for the Maximum Pressure Ratio

Net work as a function of pressure ratio. Let $r \equiv P_2/P_1$ and $a \equiv \frac{\gamma-1}{\gamma}$.

Using $T_2 = T_1 r^a$ and $T_4 = T_3 r^{-a}$,

$$w_{\text{net}}(r) = c_p [(T_3 - T_4) - (T_2 - T_1)] = c_p [(T_3 - T_1) - T_3 r^{-a} - T_1 r^a]. \quad (8)$$

Differentiate and set to zero.

$$\begin{aligned}
\frac{dw_{\text{net}}}{dr} &= c_p \left[-T_3 \frac{d}{dr}(r^{-a}) - T_1 \frac{d}{dr}(r^a) \right] \\
&= c_p [-T_3(-a)r^{-a-1} - T_1(a)r^{a-1}] \\
&= c_p a [T_3 r^{-a-1} - T_1 r^{a-1}] = 0.
\end{aligned} \tag{9}$$

Solve for r . From (9):

$$T_3 r^{-a-1} = T_1 r^{a-1} \implies T_3 = T_1 r^{2a} \implies r^{2a} = \frac{T_3}{T_1}.$$

Therefore,

$$r^* \equiv r_{\max w} = \left(\frac{T_3}{T_1} \right)^{\frac{1}{2a}} = \left(\frac{T_3}{T_1} \right)^{\frac{\gamma}{2(\gamma-1)}}. \tag{10}$$

Result: Therefore, the compressor pressure ratio that maximizes Brayton-cycle net specific work is

$$r_p^* = \left(\frac{T_3}{T_1} \right)^{\frac{\gamma}{2(\gamma-1)}} = \left(\frac{1523.15}{303.15} \right)^{\frac{1.4}{2(0.4)}} = (5.0244)^{1.75} \approx \boxed{16.86}.$$

States at r^* .

$$T_2^* = T_1 (r^*)^a, \quad T_4^* = T_3 (r^*)^{-a}.$$

1.5 Minimum Air Flow for 100 MW in (MMSCFD)

Use the optimum pressure ratio. At r_p^* (maximizes w_{net}), the required mass flow for a target power is minimized.

State temperatures and specific work. Let $a \equiv \frac{\gamma-1}{\gamma} = \frac{0.4}{1.4} = 0.285714$.

With $r_p = 16.86$,

$$T_2 = T_1 r_p^a = 303.15 \text{ K } (16.86)^{0.285714} = 679.5 \text{ K},$$

$$T_4 = \frac{T_3}{r_p^a} = \frac{1523.15 \text{ K}}{(16.86)^{0.285714}} = 679.5 \text{ K}.$$

Specific work (per unit mass of air).

$$w_c = c_p (T_2 - T_1), \tag{11}$$

$$w_t = c_p (T_4 - T_3), \tag{12}$$

$$\begin{aligned}
w_{\text{net}} &= w_c + w_t = c_p [(T_2 - T_1) + (T_4 - T_3)] \\
&= -c_p [(T_3 - T_4) - (T_2 - T_1)].
\end{aligned} \tag{13}$$

Numerical evaluation

$$w_c = 1.004 \text{ kJ kg}^{-1} \text{ K}^{-1} (679.5 - 303.15) \text{ K} = 377.86 \text{ kJ kg}^{-1},$$

$$w_t = -1.004 \text{ kJ kg}^{-1} \text{ K}^{-1} (1523.15 - 679.5) \text{ K} = -847.03 \text{ kJ kg}^{-1},$$

$$w_{\text{net}} = w_c + w_t = 377.86 - 847.03 = \boxed{-469.17 \text{ kJ kg}^{-1}}.$$

Interpretation: $w_{\text{net}} < 0$ indicates net shaft **output** from the turbine–compressor pair (work leaving the control volume). For sizing to 100 MW net output, use the magnitude $|w_{\text{net}}| = 469.17 \text{ kJ kg}^{-1}$ in $\dot{m} = \dot{W}_{\text{net}}/|w_{\text{net}}|$. (assisted by ChatGPT, OpenAI, 2025)

Mass flow for 100 MW.

$$\dot{W}_{\text{net}} = \dot{m} w_{\text{net}} \Rightarrow \dot{m} = \frac{100 \text{ MW}}{469.17 \text{ kJ kg}^{-1}} = \frac{100\,000 \text{ kJ s}^{-1}}{469.17 \text{ kJ kg}^{-1}} = \boxed{213.14 \text{ kg s}^{-1}}.$$

Convert to MMSCFD (standard)(assisted by ChatGPT, OpenAI, 2025):

$P_{\text{std}} = 1 \text{ atm} = 101.3 \text{ kPa}$, $T_{\text{std}} = 288.706 \text{ K}$. Gas constant for air:

$$R = \frac{8.314}{28.97} \frac{\text{kJ}}{\text{kmol K}} \bigg/ \frac{\text{kg}}{\text{kmol}} \approx \boxed{0.287 \text{ kJ kg}^{-1} \text{ K}^{-1}}.$$

Standard density (ideal gas):

$$\rho_{\text{std}} = \frac{P_{\text{std}}}{RT_{\text{std}}} = \frac{101.3 \text{ kPa}}{0.287 \text{ kPa m}^3 \text{ kg}^{-1} \text{ K}^{-1} \cdot 288.706 \text{ K}} \approx \boxed{1.223 \text{ kg m}^{-3}}.$$

Unit conversions:

$$1 \text{ MMSCFD} = 10^6 \frac{\text{ft}^3}{\text{day}} \times \frac{0.0283 \text{ m}^3}{\text{ft}^3} \times \frac{\text{day}}{86\,400 \text{ s}} \times \rho_{\text{std}} \approx \boxed{0.4001 \text{ kg s}^{-1}}.$$

Therefore,

$$\text{MMSCFD} = \frac{\dot{m}}{0.4001 \text{ kg s}^{-1} \text{ per MMSCFD}} = \frac{213.14 \text{ kg s}^{-1}}{0.4001 \text{ kg s}^{-1}} = \boxed{532.72 \text{ MMSCFD}}.$$

Result: The minimum air flow required to produce 100 MW in the optimal pressure ratio is $\boxed{532.72 \text{ MMSCFD}}$ (based on the stated SCF).

1.6 Cycle Efficiency at Maximum Power

Definition: The Brayton thermal efficiency is the ratio of net output to heat input (magnitudes):

$$\eta = \frac{|w_{\text{net}}|}{q_{\text{in}}} = \frac{(T_3 - T_4) - (T_2 - T_1)}{T_3 - T_2}. \quad (14)$$

At the maximum-power pressure ratio. From §1.4, the optimal ratio satisfies r_p^* with

$$T_2^* = T_1 (r_p^*)^{\frac{\gamma-1}{\gamma}} = \frac{T_3}{(r_p^*)^{\frac{\gamma-1}{\gamma}}} = T_4^* = \sqrt{T_1 T_3}.$$

Substitute $T_2^* = T_4^* = \sqrt{T_1 T_3}$ into (14):

$$\eta_{\max \text{ power}} = 1 - \frac{T_4^* - T_1}{T_3 - T_2^*} = 1 - \sqrt{\frac{T_1}{T_3}}. \quad (15)$$

Numerical value With $T_1 = 303.15 \text{ K}$ and $T_3 = 1523.15 \text{ K}$,

$$\eta_{\max \text{ power}} = 1 - \sqrt{\frac{303.15}{1523.15}} = 1 - 0.4462 = \boxed{0.5538} \quad (\approx \boxed{55.38 \%}).$$

Consistency check (using c_p values). Using $T_2^* = T_4^* \approx 679.5 \text{ K}$,

$$|w_{\text{net}}| = c_p [(T_3 - T_4^*) - (T_2^* - T_1)] \approx 469.17 \text{ kJ kg}^{-1}, \quad q_{\text{in}} = c_p (T_3 - T_2^*) \approx 847.03 \text{ kJ kg}^{-1},$$

so $\eta = |w_{\text{net}}|/q_{\text{in}} \approx 469.17/847.03 \approx 0.5538$, matching (15).

Result:

$$\eta_{\max \text{ power}} \approx 55.38 \%.$$

1.7 Pressure Ratio that Maximizes Cycle Efficiency

With $a = \frac{\gamma-1}{\gamma} = 0.285714$, $T_2 = T_1 r_p^a$, $T_4 = T_3/r_p^a$, and

$$\eta = 1 - \frac{q_{\text{out}}}{q_{\text{in}}} = 1 - \frac{c_p(T_4 - T_1)}{c_p(T_3 - T_2)} = 1 - \frac{T_3/r_p^a - T_1}{T_3 - T_1 r_p^a} = 1 - \frac{1}{r_p^a}.$$

Since heat addition from $2 \rightarrow 3$ is isobaric, $T_2 < T_3$; thus η increases with r_p and is maximized as $T_2 \rightarrow T_3$. From $T_2/T_1 = r_p^a$, the limiting (maximum) pressure ratio is

$$r_{p,\max \eta}^a = \frac{T_3}{T_1} \Rightarrow r_{p,\max \eta} = \left(\frac{T_3}{T_1}\right)^{1/a} = \left(\frac{T_3}{T_1}\right)^{\frac{\gamma}{\gamma-1}}.$$

Result:

$$r_{p,\max \eta} = \left(\frac{1523.15}{303.15}\right)^{3.5} \approx \boxed{284.31}.$$

1.8 Maximum Cycle Efficiency

Using $\eta_{\max} = 1 - \frac{1}{r_{\max}^a}$ with $a = \frac{\gamma - 1}{\gamma}$.

$$\eta_{\max} = 1 - \frac{1}{r_{\max}^a}$$

With $a = 0.285714$ and $r_{\max} = 284.31$,

$$\eta_{\max} = 1 - \frac{1}{(284.31)^{0.285714}} = 1 - \frac{1}{5.024} \approx \boxed{0.8010} \quad (\boxed{80.10\%}).$$

1.9 Net Power at Maximum Efficiency

At the efficiency-maximizing pressure ratio, $T_2 \rightarrow T_3$ (isobaric heat-addition end point) and, by isentropic expansion, $T_4 \rightarrow T_1$. Thus

$$q_{\text{out}} = c_p (T_4 - T_1) \rightarrow 0, \quad q_{\text{in}} = c_p (T_3 - T_2) \rightarrow 0.$$

From $\eta = 1 - \frac{q_{\text{out}}}{q_{\text{in}}}$, the ratio approaches its maximum as $q_{\text{out}} \rightarrow 0$ and *net work* is

$$w_{\text{net}} = q_{\text{in}} - q_{\text{out}} = c_p [(T_3 - T_4) - (T_2 - T_1)] \rightarrow c_p [(T_3 - T_1) - (T_3 - T_1)] = 0,$$

so the corresponding power output also vanishes.

Result:

$$\dot{W}_{\text{net}} = 0 \text{ MW (ideal limit at maximum efficiency).}$$

1.10 Carnot Efficiency of the Cycle

Result:

$$\eta_{\text{Carnot}} = 1 - \frac{T_{\min}}{T_{\max}} = 1 - \frac{T_1}{T_3} = 1 - \frac{303.15}{1523.15} \approx \boxed{0.8010} \quad (\boxed{80.10\%}).$$

1.11 Fuel (Kerosene) Required at Maximum Power

At the max-power point, use $\eta_{\max \text{ power}} = 0.5538$ and $\dot{W}_{\text{net}} = 100 \text{ MW}$.

$$\dot{Q}_{\text{in}} = \frac{\dot{W}_{\text{net}}}{\eta_{\max \text{ power}}} = \frac{100}{0.5538} \approx 180.57 \text{ MJ s}^{-1},$$

with kerosene LHV = 43 MJ kg^{-1} :

$$\dot{m}_f = \frac{\dot{Q}_{\text{in}}}{\text{LHV}} = \frac{180.57}{43} \approx \boxed{4.199 \text{ kg s}^{-1}}.$$

Convert to lb/hr:

$$\dot{m}_f = 4.199 \frac{\text{kg}}{\text{s}} \times 2.20462 \frac{\text{lb}}{\text{kg}} \times 3600 \frac{\text{s}}{\text{hr}} \approx \boxed{3.33 \times 10^4 \text{ lb/hr}} \quad (\text{about } \boxed{33,300 \text{ lb/hr}}).$$

Result:

$$\dot{m}_f \approx 4.19 \text{ kg/s or } \approx 3.33 \times 10^4 \text{ lb/hr} \quad (\text{kerosene, LHV} = 43.1 \text{ MJ kg}^{-1}).$$

Final answers

1. **Pressure ratio for max net power:** $r_p^* = 16.86$.
2. **Minimum air flow for 100 MW:** $\dot{m}_{\text{air}} = 213.14 \text{ kg/s} = 532.72 \text{ MMSCFD}$
(1 atm, 60°F basis).
3. **Efficiency at max power:** 55.38% .
4. **Pressure ratio for max efficiency:** $r_{p,\text{max } \eta} = 284.31$.
5. **Maximum efficiency:** 80.10% .
6. **Net power at max efficiency:** 0 MW (ideal limit).
7. **Carnot efficiency (between T_1 and T_3):** 80.10% .
8. **Kerosene required at 100 MW, $\eta_{\text{max power}}$:** $\dot{m}_f = 4.19 \text{ kg/s} \approx 3.33 \times 10^4 \text{ lb/hr}$.

2 Problem B: Vapor–Fraction of a Two-Phase Mixture

2.1 Known Data and Setup

We are given the molar composition of the mixture z_i for $i = 1, \dots, 14$ with $\sum_i z_i = 1$, and the thermodynamic values related to the physical properties of the hydrocarbon mixture.

zi	Ki
0.117598	49.905777
0.002775	13.445753
0.003435	3.474914
0.003171	1.282547
0.001189	0.628198
0.000793	0.477677
0.033971	0.064418
0.044667	0.055577
0.002572	0.226865
0.065649	0.183574
0.412276	0.092707
0.032581	0.136715
0.225623	0.072252
0.0537	0.038234

Figure 2: Given z_i and K_i values.

2.2 Vapor Fraction Equation

$$F(\alpha) = \sum_{i=1}^{14} \frac{z_i (K_i - 1)}{1 + \alpha (K_i - 1)} = 0. \quad (16)$$

2.3 Derivative

Starting from

$$F(\alpha) = \sum_{i=1}^{14} \frac{z_i (K_i - 1)}{1 + \alpha (K_i - 1)} = \sum_{i=1}^{14} f_i(\alpha),$$

differentiate term-by-term. For a fixed i , set

$$f_i(\alpha) = \frac{z_i (K_i - 1)}{1 + \alpha (K_i - 1)} = z_i (K_i - 1) [1 + \alpha (K_i - 1)]^{-1}.$$

Let $g_i(\alpha) = 1 + \alpha (K_i - 1)$. Then $g'_i(\alpha) = K_i - 1$ and

$$\frac{d}{d\alpha} g_i(\alpha)^{-1} = -g'_i(\alpha) g_i(\alpha)^{-2} = -(K_i - 1) [1 + \alpha (K_i - 1)]^{-2}.$$

By the chain rule,

$$\begin{aligned} \frac{df_i}{d\alpha} &= z_i (K_i - 1) \frac{d}{d\alpha} (g_i(\alpha)^{-1}) \\ &= z_i (K_i - 1) \left(- (K_i - 1) [1 + \alpha (K_i - 1)]^{-2} \right) \\ &= - \frac{z_i (K_i - 1)^2}{[1 + \alpha (K_i - 1)]^2}. \end{aligned}$$

Summing over i gives

$$\frac{dF}{d\alpha} = - \sum_{i=1}^{14} \frac{z_i (K_i - 1)^2}{[1 + \alpha (K_i - 1)]^2} \quad (17)$$

2.4 Methods Used

We compute α using:

- **Newton's Method:**

$$\alpha^{(k+1)} = \alpha^{(k)} - \frac{F(\alpha^{(k)})}{F'(\alpha^{(k)})}, \quad k = 0, 1, 2, \dots$$

- **Secant Method:**

$$\alpha^{(k+1)} = \alpha^{(k)} - F(\alpha^{(k)}) \frac{\alpha^{(k)} - \alpha^{(k-1)}}{F(\alpha^{(k)}) - F(\alpha^{(k-1)})}, \quad k = 1, 2, \dots$$

k	ak-1	ak	F(ak)	F(ak-1)	ak+1
1	0	1	-9.9222097	5.0015963	0.3351421
2	1	0.3351421	-0.7996945	-9.9222097	0.2768596
3	0.3351421	0.2768596	-0.6521374	-0.7996945	0.0192766
4	0.2768596	0.0192766	2.1894918	-0.6521374	0.2177458
5	0.0192766	0.2177458	-0.4805828	2.1894918	0.1820236
6	0.2177458	0.1820236	-0.3532231	-0.4805828	0.0829507
7	0.1820236	0.0829507	0.3038302	-0.3532231	0.1287633
8	0.0829507	0.1287633	-0.0903016	0.3038302	0.118267
9	0.1287633	0.118267	-0.0201945	-0.0903016	0.1152435
10	0.118267	0.1152435	0.0016883	-0.0201945	0.1154767
11	0.1152435	0.1154767	-2.957E-05	0.0016883	0.1154727
12	0.1154767	0.1154727	-4.27E-08	-2.957E-05	0.1154727
13	0.1154727	0.1154727	1.081E-12	-4.27E-08	0.1154727
14	0.1154727	0.1154727	1.665E-16	1.081E-12	0.1154727
15	0.1154727	0.1154727	-1.11E-16	1.665E-16	0.1154727
16	0.1154727	0.1154727	0	-1.11E-16	0.1154727
17	0.1154727	0.1154727	0	0	#DIV/0!

(a) Secant method.

k	ak	F(ak)	F'(ak)
0	0.5	-1.2232877	-2.8965304
1	0.0776714	0.3701138	-13.178454
2	0.1057562	0.0761692	-8.3623851
3	0.1148647	0.0044883	-7.4105718
4	0.1154704	1.714E-05	-7.3540927
5	0.1154727	2.516E-10	-7.3538769
6	0.1154727	2.776E-16	-7.3538769
7	0.1154727	-1.11E-16	-7.3538769
8	0.1154727	0	-7.3538769
9	0.1154727	0	-7.3538769
10	0.1154727	0	-7.3538769
11	0.1154727	0	-7.3538769
12	0.1154727	0	-7.3538769
13	0.1154727	0	-7.3538769
14	0.1154727	0	-7.3538769

(b) Newton's method.

Figure 3: Iteration tables.

Results:

Newton

$$\alpha^{(0)} = 0.5$$

$$\alpha^* \approx 0.1154727$$

Secant

$$\alpha^{(0)} = 0, \alpha^{(1)} = 1$$

$$\alpha^* \approx 0.1154727$$

2.5 Verification with Excel Goal Seek

As a cross-check, Excel's Goal Seek was used to set $F(\alpha) = 0$ by changing α . It converged to $\alpha \approx 0.1155277$, which agrees with the Newton/Secant root $\alpha \approx 0.1154727$.

Solution ▾	a ▾	F(a) ▾
initial guess	0.2	-0.4205025
goal seek	0.1155277	-0.0004043

Figure 4: Excel Goal Seek confirmation of the vapor fraction: $F(\alpha) = 0$ gives $\alpha \approx 0.1155277$.

3 Conclusions

Problem A — Ideal Brayton cycle.

- The pressure ratio that maximizes net specific work is $r_p^* \approx 16.86$.
- At r_p^* , the cycle efficiency is $\eta \approx 55.38\%$ and the minimum air flow to produce 100 MW is $\dot{m}_{\text{air}} \approx 213.14 \text{ kg/s}$ ($\approx 532.72 \text{ MMSCFD}$ on the stated SCF basis).
- The pressure ratio that maximizes efficiency is $r_{p,\text{max } \eta} \approx 284.31$; the corresponding maximum efficiency is $\eta_{\text{max}} \approx 80.10\%$ (Carnot between T_1 and T_3).
- At maximum efficiency, the ideal-cycle net power tends to 0 MW (since $T_2 \rightarrow T_3$ and $T_4 \rightarrow T_1$).
- For 100 MW at the max-power point, the fuel requirement is $\dot{m}_f \approx 4.19 \text{ kg/s}$ ($\approx 3.33 \times 10^4 \text{ lb/hr}$) using LHV(kerosene) $\approx 43.1 \text{ MJ/kg}$.

Problem B — Vapor-Fraction Equation.

- The vapor equation $F(\alpha) = \sum_i \frac{z_i(K_i - 1)}{1 + \alpha(K_i - 1)} = 0$ has a root in for the given data.
- Newton and Secant methods both converged to the same vapor fraction, $\alpha \approx 0.1154727$.
- Excel Goal Seek confirmed the root computed from the converged α .